

Equipment & Connectivity

Management Platform

ECM

Systems Analysis & Design

Daniel Martínez Gil · Dylan Silva Orrego · Juan Triana Paipa

Computer Engineering Program · Facultad de Ingeniería

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Problem Statement

Manual Resource Management

Equipment loans tracked via spreadsheets or paper forms, causing frequent data inconsistencies.

Lack of Traceability

No centralized system tracks device lifecycle states: available, assigned, overdue, or in repair.

Risk of Asset Loss

80% of surveyed students received a device in poor condition; no automated recovery mechanism existed.

Student Connectivity Gaps

40% of students could not take devices off campus, directly widening the digital divide.

Target population: ~7,000 students · Estimated equipment demand: 12% (~840 requests/cycle) · Prior SUS score: below threshold

Project Objectives

General Objective

Design and implement a centralized, automated platform to manage the full lifecycle of institutional technological assets and connectivity resources equitably for up to 7,000 users.

- 1 Automate resource allocation via weighted Scoring Engine (Vulnerability · Academic Load · History)
- 2 Achieve $\geq 98\%$ asset recovery rate through automated Academic Hold integration
- 3 Increase asset rotation efficiency by 15% compared to manual tracking baseline
- 4 Deliver 99.9% uptime SLA with fail-safe ISP cache and real-time monitoring dashboard
- 5 Comply with ISO 9001, CMMI, IEEE, ISO 31000, and PMBOK quality & PM standards

Systems Analysis Findings

Stakeholders

- Students — End users; request resources
- Administrators — Validate, allocate, maintain
- Donors/Lenders — Contribute & monitor impact

Survey Key Findings

40%

Could not take
devices off campus

80%

Received device
in poor condition

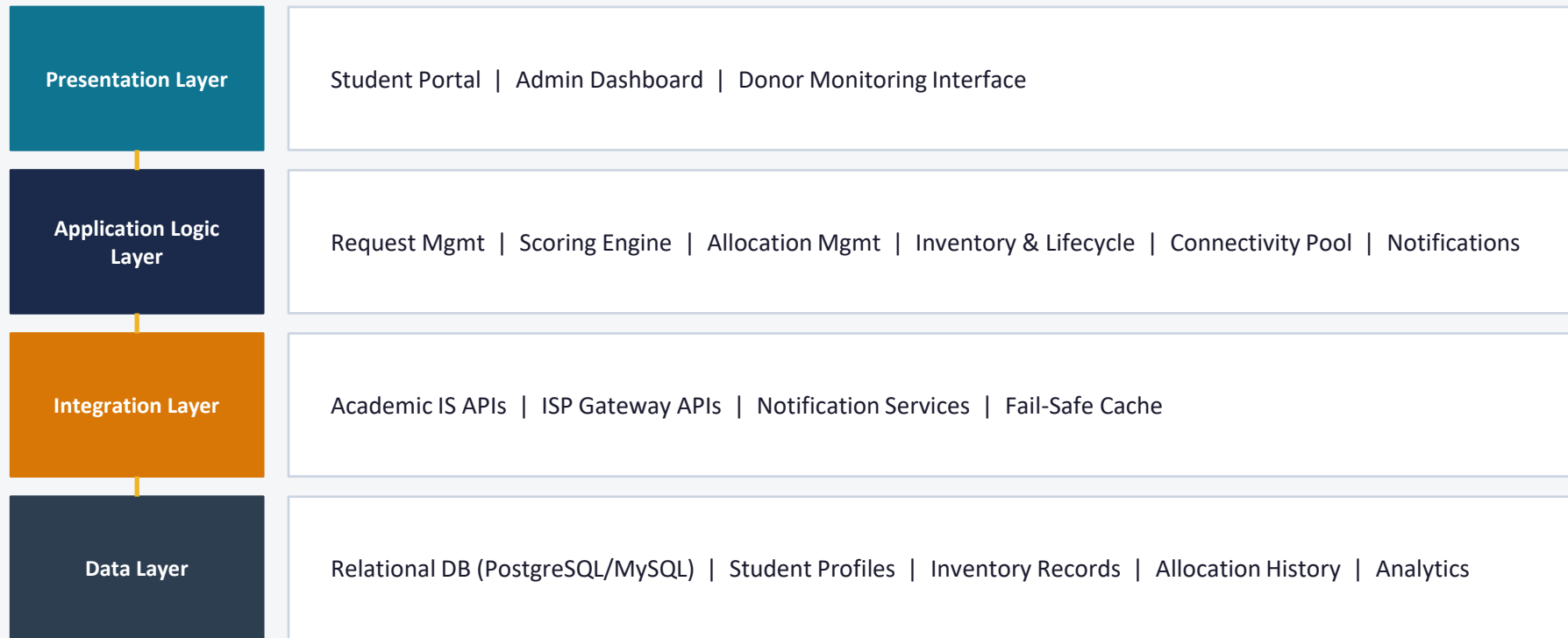
60%

Rated service
regular or below

Patterns Detected

- Nonlinear demand peaks during exam periods (+67%)
- Maintenance cascade risk when multiple devices fail simultaneously
- Connectivity circumvention: ~12% bypass attempts
- First-come model disadvantages vulnerable students

System Architecture — Layered Design



Clean Architecture principles (ISO/IEC 25010) ensure loose coupling, high cohesion, and separation of concerns across all four layers.

Core Functional Modules

Request Management

Validates student documentation; routes approved requests to the Scoring Engine for priority calculation.

Automated Scoring Engine

Computes $S = (V \cdot 0.5) + (A \cdot 0.3) + (H \cdot 0.2)$; recalibrates weights monthly based on demand/supply ratio.

Allocation Management

Matches scored requests to available inventory; maintains waiting list with dynamic prioritization.

Inventory Management

Tracks asset states: Available → Assigned → In-Repair → Overdue → Retired with full audit log.

Connectivity Pool Manager

Manages routers and data plans; enforces fair-use policies with domain filtering for academic traffic.

Notification & Alert System

Automated reminders for returns, Academic Hold triggers, overdue alerts via Slack/Email channels.

Robust Design Principles

Scalability

CMMI L3

Modular services support horizontal auto-scaling; tested up to 1,400 concurrent requests.

Reliability

IEEE 730

99.9% uptime SLA; ISP Fail-Safe Cache prevents service loss during outages.

Maintainability

ISO 25010

Clean Architecture decouples business logic; enables independent module updates.

Modularity

CMMI L2

Strategy & Observer design patterns isolate allocation algorithms and notification logic.

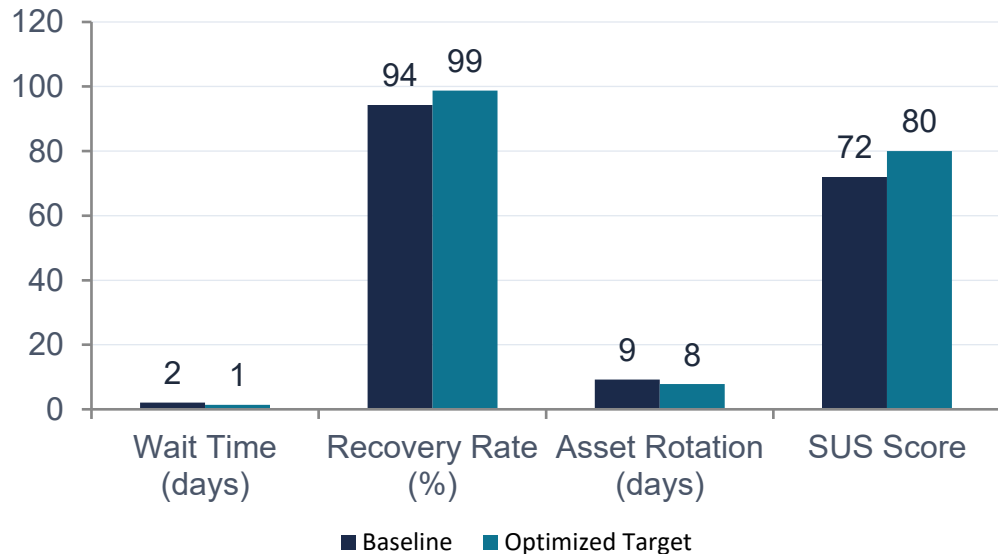
Fault Tolerance

ISO 31000

Automated state transitions (In-Repair, Academic Hold) contain failure propagation.

Quality Assurance Framework — ISO 9001 & KPIs

KPI: Baseline vs Optimized



QA Methods

ISO 9001

Process documentation, continuous improvement cycles

UAT

Pilot group completes request flow in under 3 minutes

Audit Logging

Every asset state change timestamped & signed

Health Checks

ISP heartbeat monitors; admin live dashboard

Acceptance Criteria

SUS ≥ 80 ; recovery $\geq 98\%$; uptime $\geq 99.9\%$

Data Integrity Validation: periodic automated sanity checks ensure every 'Assigned' device is linked to an active student record with no orphaned entries.

Risk Management Matrix — ISO 31000 / PMBOK

Risk	Impact	Probability	Mitigation Strategy
Peak demand overload (exam periods)	HIGH	HIGH	Auto-scaling at 800 req/min; dynamic prioritization with waiting-list manager
ISP gateway failure / connectivity loss	HIGH	MED	Fail-Safe Cache provides emergency tokens; heartbeat health checks trigger alerts
Maintenance avalanche (>25% fleet in repair)	HIGH	MED	Staggered preventive maintenance: $\leq 10\%$ fleet in repair at any time
Data security / privacy breach (FERPA / institutional policy)	HIGH	LOW	Anonymized socioeconomic data; role-based access control; full audit logging
Low user adoption / resistance to change	MED	MED	UAT pilot, onboarding training, < 3-minute request UX target (SUS ≥ 80)

Project Management Plan — PMBOK Framework

Team Structure & Roles

Daniel Martínez Gil

Project Lead & Systems Architect

Dylan Silva Orrego

Backend Dev & Integration Engineer

Juan Ramírez Hernández

QA Lead & Risk Manager

Juan Triana Paipa

Frontend Dev & UX Specialist

Methodology

- PMBOK 7 — Scope, Schedule, Risk, Quality, and Communications management areas
- Incremental SDLC — Phased delivery with validation gates between phases
- Agile-inspired sprints within phases for continuous feedback integration

Communication & Control Plan

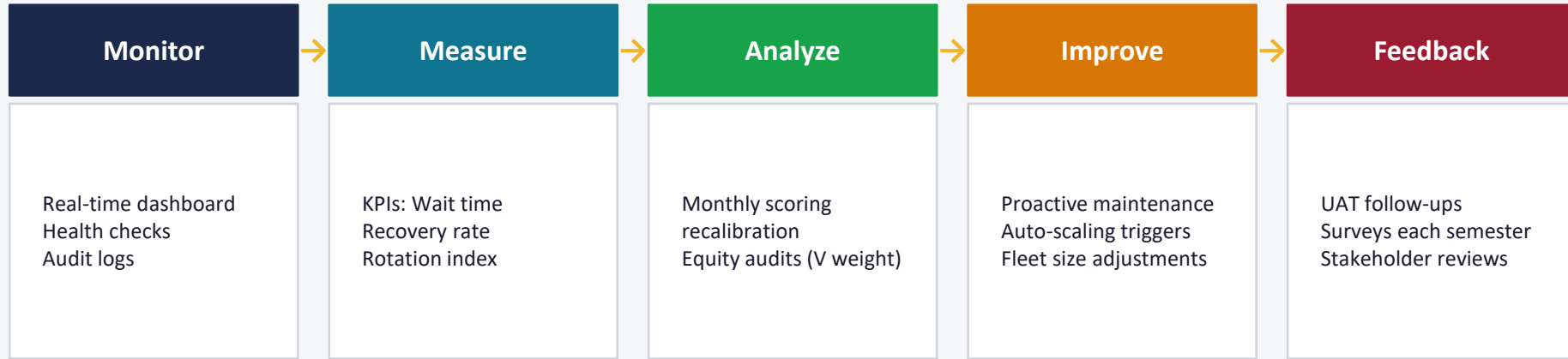
- Weekly sprint review with professor/stakeholders
- GitHub repository — all artifacts versioned per workshop
- Risk register updated before each phase gate
- Slack/Email automated alerts for critical system events

Implementation Roadmap — Phased Deployment

Phase 01 Core MVP	Phase 02 Scoring Engine + Integrations	Phase 03 Pilot & Stress Testing	Phase 04 Full Deployment
Weeks 1–4	Weeks 5–9	Weeks 10–13	Weeks 14–16
1 Inventory management module 2 Basic request submission forms 3 PostgreSQL schema & REST API 4 Student & Admin portal scaffolding	1 Automated Scoring Engine $S=(V \cdot A \cdot H)$ 2 ISP API integration & Fail-Safe Cache 3 Academic IS integration (Academic Hold) 4 Notification & alert system	1 Controlled rollout: sample user group 2 UAT — request flow under 3 min ($SUS \geq 80$) 3 Security & data privacy audit 4 Simulation validation (DES + ABM)	1 Full institutional deployment (7,000 users) 2 Live monitoring dashboard activation 3 Training & user onboarding 4 Continuous improvement feedback loops

► Analysis → Design → Robust Design → Simulation → Pilot → Full Deployment

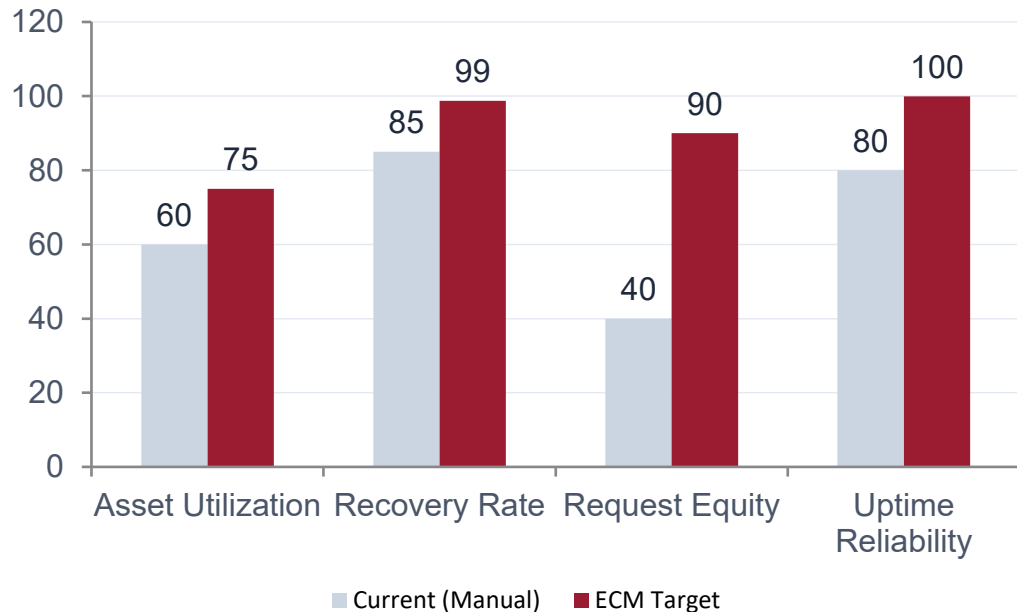
Continuous Improvement & System Evolution



Future Evolution Roadmap

- Multi-semester ABM simulation to model long-term compliance behavior evolution
- Real device degradation curves from maintenance logs replace simulated functions
- Donor contribution stochastic modeling for inventory growth scenario planning
- Predictive demand forecasting module using historical allocation cycle data
- Wait-time bonus parameter for students in queue beyond 5 days (equity safeguard)

Performance: Manual vs ECM (%)



+15%

Asset Rotation Efficiency

Devices assigned faster; fewer idle assets in inventory.

98.7%

Recovery Rate (from 85%)

Academic Hold mechanism enforces timely returns.

34%

Wait Time Reduction (DES)

Scoring Engine vs. first-come: 2.1 → 1.4 days.

0

Manual Tracking Errors

Full automation replaces spreadsheet-based processes.

Conclusions



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Workshop 1

Systems Analysis — Stakeholders, survey data, emergent behavior patterns, critical assessment of manual processes.

Workshop 2

System Design — Layered architecture, Scoring Engine formula, Connectivity Pool, implementation roadmap.

Workshop 3

Robust Design — ISO 9001/CMMI/IEEE standards, risk matrix (ISO 31000), PMBOK project plan, quality KPIs.

Workshop 4

Simulation — DES + ABM validated 4/5 design decisions; revealed maintenance avalanche and scaling ceiling.

ECM is a feasible, equitable, and technically validated solution